



AMERICAN UNIVERSITY OF PHNOM PENH

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COSC 251 Fall 2026: Assignment 1 – Arrays (50 points)

Deadline: Sep 30, 11:59 PM

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Overview: For this assignment, you are required to implement two versions of an array class: an `UnorderedArray` and an `OrderedArray`. Both classes should provide methods for insertion, deletion, search, retrieval, size, and resizing.

Be sure to consider the time complexity of each method to receive full credit. Review Chapter 2 and ensure you understand the differences between an unordered array and an ordered array.

Project Requirements:

1. Both classes should include an integer array `Integer[] arr` as a member variable. Elements may be set to `null` to indicate unused positions. Additional member variables may be added if necessary.
2. Both classes should have a constructor that accepts an integer parameter used to initialize the size of `arr[]`.
3. Both classes should implement an `insert(int x)` method to insert an integer into `arr[]`. If the array is full, the method should automatically resize the array.
4. Both classes should implement a `delete(int x)` method that removes the first occurrence of an integer from `arr[]` if it exists and returns `true`. If the integer does not exist, the method should return `false`. After deletion, remaining elements should shift left so that all non-null elements remain contiguous.
5. Both classes should implement a `find(int x)` method that searches for an integer in `arr[]`. If the integer exists, return its index; otherwise, return `-1`.
6. Both classes should implement a `get(int index)` method that returns the integer at the given index in `arr[]`. If the index is out of bounds, the method should throw an `IndexOutOfBoundsException`. The method may return `null` if no element exists at that position.
7. Both classes should implement a `size()` method that returns the total capacity of `arr[]`.
8. Both classes should implement a `count()` method that returns the number of non-null elements currently stored in `arr[]`.
9. Both classes should implement a `resize(int newSize)` method that changes the size of `arr[]` to `newSize`. Existing elements and their order should be preserved as much as possible. Any elements beyond the new size should be discarded.
10. In `UnorderedArray`, `arr[]` remains unsorted. In `OrderedArray`, `arr[]` must always remain sorted in ascending order.
11. Each method must include a short comment specifying and explaining its time complexity.

What to Turn In: This is a group assignment; each group should submit only once.

1. Your `UnorderedArray` Java class with all required methods and functionality. The file must be named `UnorderedArray.java`, and the class must be named `UnorderedArray`.
2. Your `OrderedArray` Java class with all required methods and functionality. The file must be named `OrderedArray.java`, and the class must be named `OrderedArray`.
3. Optional: any additional supporting Java classes.
4. Note: Do not submit a `main` class. The submitted classes must not include a `main` method.

Grading:

1. Program Design (10): The overall quality of your solution design.
2. Program Execution (15): The program runs correctly without errors.
3. Specification Satisfaction (15): The program fully meets the given requirements.
4. Coding Style and Readability (10): Code is well-formatted, clear, and includes meaningful comments.